Learning Objective 1.1: I can explain what an antenna is, describe its role in a wireless system, and recognize common antenna types by sight.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Name that antenna.** Each description below is what you would see if you walked up to the hardware. For each, name the antenna family and give its *job* — the reason an engineer reaches for that shape — in one or two sentences.

   (a) **(i)** A straight metal rod, split and fed at its centre, roughly half a wavelength long overall. **(ii)** A single rod about a quarter of a wavelength tall, standing on a large sheet of metal or a vehicle roof and fed at its base.

   (b) A thin rectangle of copper etched on a grounded dielectric board, about half a wavelength across, fed from underneath or from a trace at its edge.

   (c) A rectangular metal pipe that flares open at one end, fed from a waveguide at the other.

   (d) **(i)** A row of parallel rods of slightly different lengths mounted along a boom, with only one of them connected to a cable. **(ii)** A large curved metal dish with a small antenna suspended at its focus.

2. **What an antenna is, and what it is for.** Answer each part in one or two complete sentences.

   (a) Define an antenna as a *transducer*. Name the two kinds of wave it sits between, and say which component in an RF link block diagram (source → amplifier → transmission line → antenna → channel) is the only one that touches free space.

   (b) Antennas do not create energy. Given that, explain what a vendor means by "this antenna has 30 dBi of gain."

   (c) You measure a tracking antenna on transmit and find a 3° beam pointed at boresight. You now use the same antenna to *receive*. State what reciprocity tells you about the receive pattern, and give two practical consequences for the receiving station.

   (d) Match each band to the mission it is usually flown for, and state in one sentence what sets the size of the antenna in each case. **Bands:** (i) 3–30MHz (HF), (ii) 225–400MHz (UHF), (iii) 1575MHz (GPS L1), (iv) 8–12GHz (X-band). **Missions:** fire-control and imaging radar; beyond-line-of-sight skywave communications; navigation; tactical manpack radio and UHF SATCOM.

Documentation: ____